

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T21 · Algebraic Fractions

Questions

Student book · questions with writing space

■ 18 pages

FAMILY MAP

Algebraic Fractions

F01	Add or subtract algebraic fractions	6 Q 17 marks
F02	Multiply or divide algebraic fractions	4 Q 16 marks
F03	Substitute or transform a rational expression	6 Q 23 marks
F04	Simplify an algebraic fraction by factor cancellation	1 Q 3 marks

Working Unit 09

4MA1-1H Q12

UNIT 1

3 marks

12 Express $\frac{5}{4} + \frac{x-3}{6x}$ as a single fraction in its simplest form.

.....
(Total for Question 12 is 3 marks)

Working Unit 25

4MA1-2H Q23

UNIT 1
4 marks

23 $\frac{4x^2 - 4x - 120}{5x^2 - 180} \div \frac{x^2 + 5x}{10x^2 + 60x} = p$ where p is an integer.

Find the value of p

Show clear algebraic working.

$$p = \dots\dots\dots$$

(Total for Question 23 is 4 marks)

Question 09

4MA1-1H Q12

MODULAR UNIT 1

3 marks

(b) Express $\frac{2x+1}{4x} + \frac{7-5x}{3x}$ as a single fraction in its simplest form.

(3)

Question 10

4MA1-1H Q11 demand

MODULAR UNIT 1

3 marks

11 Express $\frac{3}{4} + \frac{5-x}{6x}$ as a single fraction in its simplest terms.

.....

Question 31

4MA1-2H Q24 whole

MODULAR UNIT 1

5 marks

24 Solve $\frac{45x^2 - 80x}{3x^2 + x - 4} \times \left(\frac{1}{3x - 4} + \frac{1}{x} \right) = \frac{4(x + 2)}{5x - 8}$

Show clear algebraic working.

$x =$

4MA1-2H Q23 M01

4 marks

Show clear algebraic working.

Question 08

4MA1-1H Q11 Q11(b)

MODULAR UNIT 1

3 marks

(b) Express $\frac{7}{8} - \frac{x+3}{4x}$ as a single fraction in its simplest form.

.....

(3)

Question 26

4MA1-2H Q19 Q19

MODULAR UNIT 1

3 marks

19 $a = \frac{14}{3x - 7}$ $x = \frac{7}{4y - 3}$

Express a in the form $\frac{py + q}{ry + s}$ where p, q, r and s are integers.

Give your answer in its simplest form.

$a =$

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3}\right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

as a single fraction in its simplest form.

Give your answer in its simplest form.

14 (a) Solve $\frac{9a - 7}{5} - \frac{3a - 7}{4} = 4.55$

Show clear algebraic working.

$a =$
(3)

as a single fraction in its simplest form.
Show clear algebraic working.

Give your expression as a single fraction in its simplest form.

Give your answer in its simplest form.

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

13 Solve $\frac{x+3}{4} - \frac{7-x}{5} = 4.3$

Show clear algebraic working.

$x =$

(Total for Question 13 is 3 marks)

Show your working clearly.